

WORCESTER, South Africa

WMO: 688210

Lat: **33.664S**Lon: **19.419E**Elev: **199**StdP: **98.96**Time Zone: **2.00 (E02)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.1	3.5	-0.4	3.8	9.1	0.6	4.0	8.7	16.8	12.8	15.4	14.2	0.9	70	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.4	35.2	19.9	33.3	19.5	31.7	19.2	21.8	30.5	21.1	29.5	20.4	28.6	5.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.3	25.3	18.4	13.6	24.7	17.5	12.8	24.0	64.4	30.4	61.8	29.5	59.3	28.7	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.2	10.8	0.0	39.4	0.6	2.5	-0.4	41.2	-0.8	42.7	-1.1	44.1	-1.5	45.9		
(5)			-0.5	22.6	0.5	1.1	-0.9	23.4	-1.2	24.0	-1.5	24.7	-1.9	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	24.7	24.6	22.5	19.3	15.8	12.2	11.8	12.6	15.1	18.6	20.7	23.4	
	DBStd	5.40	2.57	2.51	2.92	3.17	2.48	2.41	2.56	2.39	2.71	3.30	2.92	2.78	
	HDD10.0	21	0	0	0	0	0	7	9	4	1	0	0	0	
	HDD18.3	834	0	1	3	26	86	184	204	179	102	37	12	1	
	CDD10.0	3088	455	409	388	278	179	73	65	83	154	266	322	417	
	CDD18.3	860	197	177	132	54	6	1	1	1	5	43	84	159	
	CDH23.3	6652	1616	1404	882	353	56	8	11	11	57	359	606	1290	
	CDH26.7	2599	706	605	312	103	6	1	2	1	11	119	196	538	
Wind	WSAvg	4.2	4.4	4.1	3.9	3.4	3.8	4.4	4.3	4.3	4.2	4.1	4.3	4.4	
Precipitation	PrecAvg	271	8	7	11	26	31	46	40	39	24	21	9	7	
	PrecMax	475	102	74	59	105	124	132	135	160	114	110	68	34	
	PrecMin	137	0	0	0	0	0	2	5	2	0	0	0	0	
	PrecStd	96	19	14	13	28	27	32	31	34	25	23	14	9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.5	37.3	34.7	32.9	27.5	24.5	24.6	24.9	28.3	34.4	34.0	37.3	
		MCWB	20.7	20.5	19.9	18.7	15.1	13.9	13.5	14.2	16.3	19.3	18.8	20.4	
	2%	DB	34.7	34.5	32.1	29.7	24.7	20.0	20.5	20.9	25.0	30.0	31.1	33.8	
		MCWB	20.2	19.8	19.4	18.2	15.2	12.9	13.0	13.8	15.2	17.3	17.8	19.5	
	5%	DB	32.5	32.3	30.1	27.3	22.8	18.2	18.3	18.8	22.5	27.2	29.0	31.7	
		MCWB	19.7	19.7	18.8	17.7	14.9	12.5	12.4	12.9	14.4	16.4	17.3	19.1	
	10%	DB	30.6	30.4	28.1	25.1	20.9	16.7	16.5	17.1	20.4	24.8	26.9	29.7	
		MCWB	19.2	19.3	18.3	16.9	14.1	12.0	11.8	12.2	13.5	15.7	16.7	18.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	22.7	21.8	20.1	17.4	15.0	15.2	15.9	17.5	20.3	20.5	22.3
			MCDB	32.5	31.1	29.9	28.6	23.8	20.4	21.3	21.5	24.6	31.7	29.7	32.0
2%		WB	21.7	21.7	20.6	19.2	16.3	14.1	13.8	14.6	16.1	18.5	19.4	21.1	
		MCDB	30.6	29.9	28.6	27.0	22.1	18.4	18.3	19.2	22.6	27.3	27.5	30.2	
5%		WB	20.8	21.0	19.9	18.4	15.5	13.2	12.9	13.6	15.1	17.4	18.5	20.2	
		MCDB	28.9	29.1	27.6	25.3	20.9	17.1	17.0	17.8	21.5	24.7	26.3	28.8	
10%		WB	20.0	20.2	19.1	17.6	14.8	12.5	12.1	12.6	14.0	16.4	17.6	19.4	
		MCDB	28.2	28.1	26.2	23.8	19.7	16.1	16.0	16.6	19.7	23.1	25.2	27.7	
Mean Daily Temperature Range	5% DB	MDBR	12.4	12.3	11.1	11.0	10.7	9.8	10.1	9.7	10.3	11.7	12.1	12.4	
		MCDBR	15.6	15.4	13.9	14.7	14.1	12.5	13.5	13.1	14.5	16.0	15.4	15.6	
		MCWBR	4.2	4.2	3.8	4.9	6.4	7.2	7.7	6.9	6.1	5.5	4.7	4.3	
	5% WB	MCDBR	13.8	13.3	12.2	12.9	12.1	10.0	11.1	11.3	13.3	13.8	13.6	13.9	
		MCWBR	4.2	4.2	3.8	4.6	6.0	6.2	6.7	6.2	5.9	5.1	4.7	4.2	
Clear-Sky Solar Irradiance	taub	0.341	0.341	0.335	0.317	0.309	0.300	0.303	0.327	0.338	0.353	0.336	0.341		
	taud	2.499	2.503	2.514	2.564	2.573	2.570	2.552	2.477	2.415	2.396	2.486	2.492		
	Ebn at Noon	997	976	945	909	863	845	860	883	926	952	995	1000		
	Edh at Noon	114	110	101	86	75	71	75	92	110	121	115	116		
All-Sky Solar Radiation	RadAvg	8.25	7.42	6.01	4.37	3.12	2.63	2.92	3.67	5.05	6.51	7.71	8.30		
	RadStd	0.25	0.20	0.23	0.22	0.19	0.17	0.16	0.17	0.23	0.35	0.25	0.30		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	+0.36	+0.49	N/A	N/A	+0.53	+0.47	-2	-74	+111

Nomenclature: See separate page